



Powered Wing's Response to Streamwise Gust Encounters

Lucas A. Duncan¹ and Jielong Cai²

University of Dayton, Dayton, Ohio, 45469, United States

Sidaard Gunasekaran³

University of Dayton, Dayton, Ohio, 45469, United States

We continue our study of a powered wing's response to streamwise gusts and investigate its performance. A full span aspect ratio 2.67 wing model with different propeller installation locations was subjected to streamwise gusts of varying reduced frequencies in the University of Dayton Low Speed Wind Tunnel (UD-LSWT). A range of angles of attack between -10 to 20 degrees under both steady and unsteady conditions were tested. An increment in the lift coefficient and pitching moment coefficient response was found at higher reduced frequency for the powered and unpowered wings. At low angles of attack, the differences in powered and unpowered wing's responses are nearly identical across all reduced frequencies. The peak C_L and C_M occur at different phase angles at different reduced frequencies. However, at higher angles of attack, the deviation between powered and unpowered C_L increases with increase in reduced frequency. This deviation is caused by the added energy to flow around the wing by the propeller, which prevents flow separation. However, the deviations also depend on the location of the propeller along the wingspan and its advance ratio.

Nomenclature

AR	= Aspect Ratio	U_∞	= Freestream Velocity, (m/s)
C_L	= Lift Coefficient, $C_L = \frac{L}{qS_{Ref}}$	U_{avg}	= Mean Velocity, (m/s)
C_M	= Moment Coefficient, $C_M = \frac{M}{qS_{Ref}}$	α	= Angle of Attack, (Deg)
L	= Lift force, (N)	ρ	= Air density, ($\frac{kg}{m^3}$)
M	= Pitching Moment, (N.m)	k	= Reduced Frequency, $k = \frac{\omega c}{2V}$
D	= Drag force (N)	ω	= Circular Frequency, (rad/s)
T	= Thrust force (N)	c	= Chord Length, (m)
q	= Dynamic Pressure, (Pa)	σ	= Velocity Amplitude Ratio, ($\frac{V_{amp}}{\bar{V}}$)
Re	= Reynolds Number	V_{amp}	= Velocity Amplitude, (m/s)
S_{ref}	= Reference area, (m ²)	ϕ	= Phase Angle (Deg)

I. Introduction

A century of research on wings, propellers, and their combination has led to the development and successful operation of commercial air vehicles and aircraft designs. However, the rise of battery technology and the need for clean and sustainable transportation have culminated in numerous Electric Vertical Takeoff and Landing (eVTOL) air vehicles called air taxis and Advanced Air Mobility (AAM) vehicles. Drastic changes in design constraints must be

¹ Graduate Student, Department of Mechanical and Aerospace Engineering, AIAA Member. duncanl6@udayton.edu

² Graduate Student, Department of Mechanical and Aerospace Engineering, AIAA Member. caij03@udayton.edu

³ Associate Professor, Department of Mechanical and Aerospace Engineering, AIAA Member. gunasekarans1@udayton.edu

made when compared to traditional aircraft, such as Vertical Takeoff and Landing (VTOL) capability, autonomy, small footprint (compactness), energy efficiency, and a plethora of additional safety constraints when compared to conventional aircraft designs [1]. Due to these changes, the air taxi designs stray far from the current norm leading to new research avenues on system integration (wing, structure, propulsion, control, etc.). Companies like Joby Aviation, Blade Air Mobility, and Volocopter are leading the charge in eVTOL air taxi development. These vehicles operate at low altitudes and are designed to take off and land on top of buildings, necessitating VTOL capability. In order to travel effectively, the vehicle must transition from vertical to forward flight in a short period of time. Historically, this transition is the most difficult phase of flight, where unsteady conditions such as gusts can cause the aircraft to lose correct orientation and heading. Building wakes and the atmospheric boundary layer can be highly turbulent and unsteady such that any vehicle operating in such conditions can lose control, with the potential for catastrophic failure.

Micro Air Vehicles (MAVs), specifically tail-sitter configurations, have dealt with the transition between vertical and forward flight since their inception. Often, the transition is automatically stabilized by control systems set before flight, due to the complexity of pure human-input control. Hostenbach et al. [2] found part of this complexity to be before the transition even begins – hover stability becomes an issue when the MAV encounters a wind gust. Once stabilized, their MAV will decrease pitch until the stall speed is exceeded and the optimal angle of attack is reached, at which point forward flight is achieved. A sudden loss or gain in altitude is common during the transition [3][4], which can be disastrous in the low-altitude conditions that air taxis operate under. Some controllers are able to counter the instability through mitigating the sharp changes in forces and moments produced by the MAV [3]. With control, however, inaccurate gain scheduling can induce yaw oscillation when transitioning from forward to vertical flight [4]. While controllers are able to stabilize the transition, they can be flawed. A gust encounter during a controlled maneuver may bring even more instability to an already dangerous transition period.

A. Wing's Response to Streamwise Gusts

A wing's response to unsteady flow has been investigated through a variety of methods and gust profiles. Transverse, vortical, and streamwise gusts all have differing effects on the wing's force response and the flowfield around it. An overview of the three different types of gusts is discussed in Jones et al. [5]. Jones et al. state that the change in lift coefficient through small velocity amplitude ratios is due to the local dynamic pressure change. In large velocity amplitude ratios, flow separation becomes the dominating factor in how lift changes. Strangfeld et al. [7] found that the theory developed by Isaacs can match experimental results with acceptable accuracy. In their tripped boundary layer condition, the theory accurately predicted lift trends and magnitudes. However, in lower Reynolds number regimes ($<10^6$) without a boundary layer trip, the separation bubble caused a lift coefficient undershoot along with high-frequency oscillations. Based on their observations, lift could change about 30% from steady state in unsteady flows. Large changes in lift support the concern for stability in close-quarters transition flight. Similar studies were conducted at a higher Reynolds number and discovered significant deviations in experimental results compared to the theory [8]. They found the Kutta condition to be violated for their streamwise gusts, with the stagnation line on the wing changing with the surging flow.

B. Propeller's Response to Streamwise Gusts

Similar to a wing, propellers encountering gusts have been investigated. Most research has been focused on the unsteady interactions between rotors and propeller blades [9][10]. Along with the development of air taxis, the focus has shifted to propellers interacting with an external source of unsteady flow. Cai et al. [11] found that steady-state performance between different velocities could be interpolated to develop a quasi-steady model of propellers in low reduced frequency ($k \leq 0.0796$) streamwise gusts. Although some phase lag is introduced, the influence of the unsteady flow can be predicted, at least in the quasi-steady regime of flow. It was also found that propellers in forward flight in axial configuration have a different gust response when compared to edgewise flight. Others have found unsteady conditions to delay control inputs, affect power consumption, and overall performance [12][13].

C. Propeller-Wing Interactions

Propeller-wing systems are the prevailing design choice for air taxis and AAM vehicles due to their higher efficiency, lower weight, and lower cost [14]. A powered wing will behave differently than an unpowered wing, due to propeller-wing interactions. A plethora of literature exists on active flow control through blown wings that are used to augment the lift coefficient by passing the engine wake on top of the wing, energizing the flow. Propellers will

cause local upwash on the side of blade upward rotation, and a subsequent local downwash on its blade downward motion [15]. Energy transferred to the flow field from the propeller also increases the lift generated by the wing. Srivathsan and Rauleder [16] found that propellers delay separation on the upper surface of the wing, with an increased lift curve slope. The optimal downstream distance for the propeller to be located is one propeller radius away from the wing, implying that optimal configurations for a propeller-wing system can be reached using a number of design considerations.

Almost no literature currently exists that explores the behavior of a powered wing to different types of gusts. The added thrust component, and the interactions between the propeller wake and the wing's boundary layer can cause additional changes in the wing's response. As such, the aim of the current research is to determine the gust response of a standalone and a powered wing model and compare it to the theoretical predictions from Isaacs' theory [6], and characterize the role of thrust and propeller-wing interaction in the wing's response to streamwise gusts. Once the influence of the propeller on the wing is understood in streamwise gusts, more complex studies can be conducted to apply this knowledge to ideal and perturbed transition flight.

II. Experimental Setup

All experiments were conducted in the University of Dayton Low Speed Wind Tunnel (UD-LSWT). The tunnel is an Eiffel type, open-jet configuration with six anti-turbulence screens, a 16:1 contraction ratio, and a 0.762m x 0.762m testing section. The testing section has a turbulence intensity of 0.1% measured by hot-wire anemometry, and is followed by a 1.370m x 1.370m collector. Placed directly after the collector is a state-of-the-art shuttering system, which has a set of rotating louvers. Changes in 1D flow velocity can be induced through systematic opening and closing of the louvers. The shuttering system was characterized to produce a sinusoidally-varying freestream at a frequency range of 0Hz to 2.25Hz [11] [17]. Figure 1 shows the open and closed configurations of the system, as well as the flow velocity change with different actuation frequencies at a wind tunnel fan rotational speed of 400 RPM. The flow velocity measurements were taken with hot-wire anemometry placed at the inlet of the testing section.

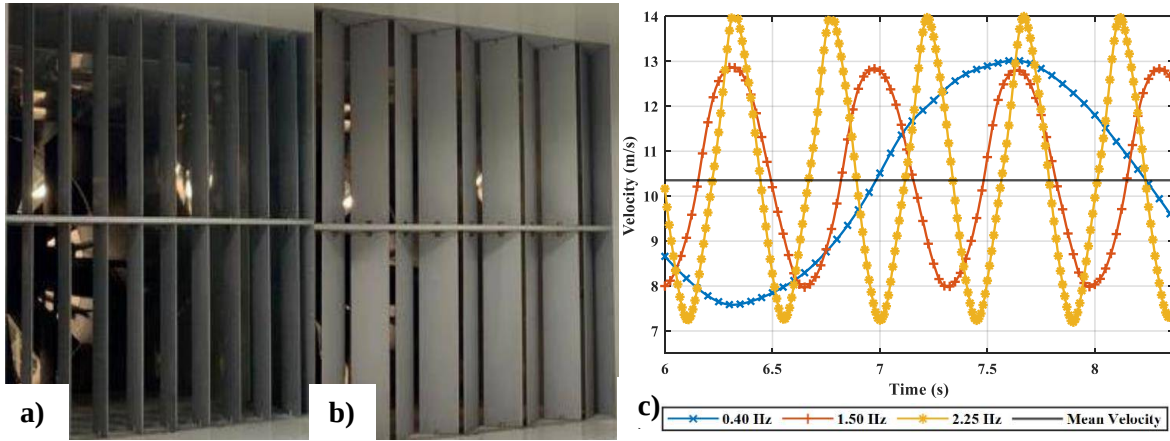


Figure 1. UD-LSWT Shuttering System a) Fully Open, b) Fully Closed, and c) Velocity Response to Actuation Frequency

The experimental setup is shown in Figure 2. The wing model has a NACA 0012 airfoil with a 0.1524m chord and a 0.4064m span, corresponding to an aspect ratio of 2.67. The design includes three modular propeller locations (midspan, semi-midspan, and wingtip) and an unpowered wing for a total of 4 configurations. The propulsion assembly consists of a Gemfan Hurricane 4024 2-blade, 0.102m diameter propeller and a T-Motor F2004 3000KV motor. The wing is mounted to an ATI Mini40-E force sensor [17], which is mounted to a streamlined support system. A rotary stage on either side of the support allows the angle of attack of the entire system to be changed. Not pictured in Figure 2 is a Dantec Dynamics type 55R01 hot-wire [18] placed upstream of the wing, used to collect instantaneous velocity measurements during actuation of the shuttering system.

The test matrix of the experiment is shown in Table 1, which includes both steady and unsteady conditions. The steady flow freestream velocity was set at 5, 7.5, and 10 m/s, where the wing's angle of attack is varied from -10 to 20° in 1° increments. The unsteady flow mean velocity corresponds to the steady freestream, with a variation of ± 1 , ± 1.5 , and ± 2 m/s respectively. The angles of attack chosen were 0°, 5°, 10°, and 15° with shuttering actuation

frequencies of 0.4, 1.5, and 2.0 Hz. These shuttering actuation frequencies correspond to reduced frequencies (k) ranging from 0.02 to 0.23 based on mean velocity and half chord length of the wing. Throttle settings were changed to achieve trimmed condition, which means that at 0° angle of attack the propeller thrust matches the drag force of the propeller-wing system. The RPM of the propeller is held constant through all cases. Tuft flow visualization with No. 6 crochet yarn was utilized to gain a visual understanding of flow on the different wing configurations. The tufts were placed in offsetting rows along the wing model, spaced about 0.0254m apart in both directions. The test matrix in Table 1 was duplicated for tuft flow visualization purposes.

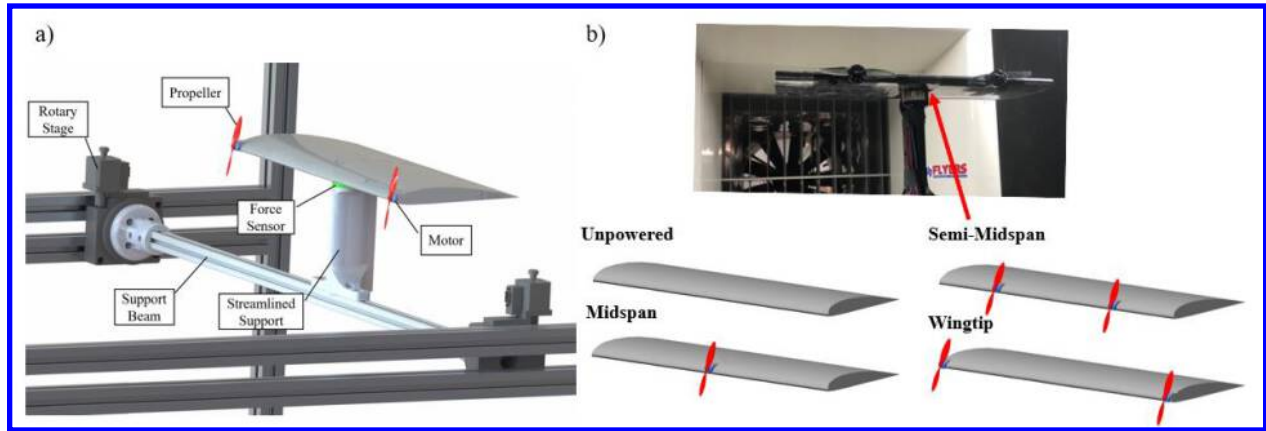


Figure 2. CAD Model for Experimental Setup of a) Wind Tunnel Testing and b) Propeller Locations

Table 1. Testing Matrix

Flow	Velocity (m/s)	Reynolds Number	Angle of Attack	Reduced Frequency (k) (Shuttering Frequency)	Configuration	Thrust Generated
Steady	5, 7.5, 10	50,000 75,000 100,000	-10° to 20°	N/A	Unpowered Midspan Semi-Midspan	$D = 0$
Unsteady	5 mean 7.5 mean 10 mean	40,000 to 110,000	0° , 5° , 10° , 15°	0.02 to 0.23 (0.4 to 2.0 Hz)	Wingtip	

A comparison must be made between the ideal theoretical velocity and the experimental velocity, as the experiment will differ slightly from the ideal sinusoidal freestream. Figure 3 below shows this comparison for all experimental conditions, where ϕ represents the phase angle. The experimental velocity profiles are an average of ~ 30 periods. At low average velocities and reduced frequencies, the two velocities match well, and the relative deviation is small. At higher amplitudes that often occur at higher velocities, the deviations from the ideal sinusoidal freestream tends to be high. The relative deviations are within reason (8% maximum) for an open jet tunnel, but must be acknowledged, nevertheless.

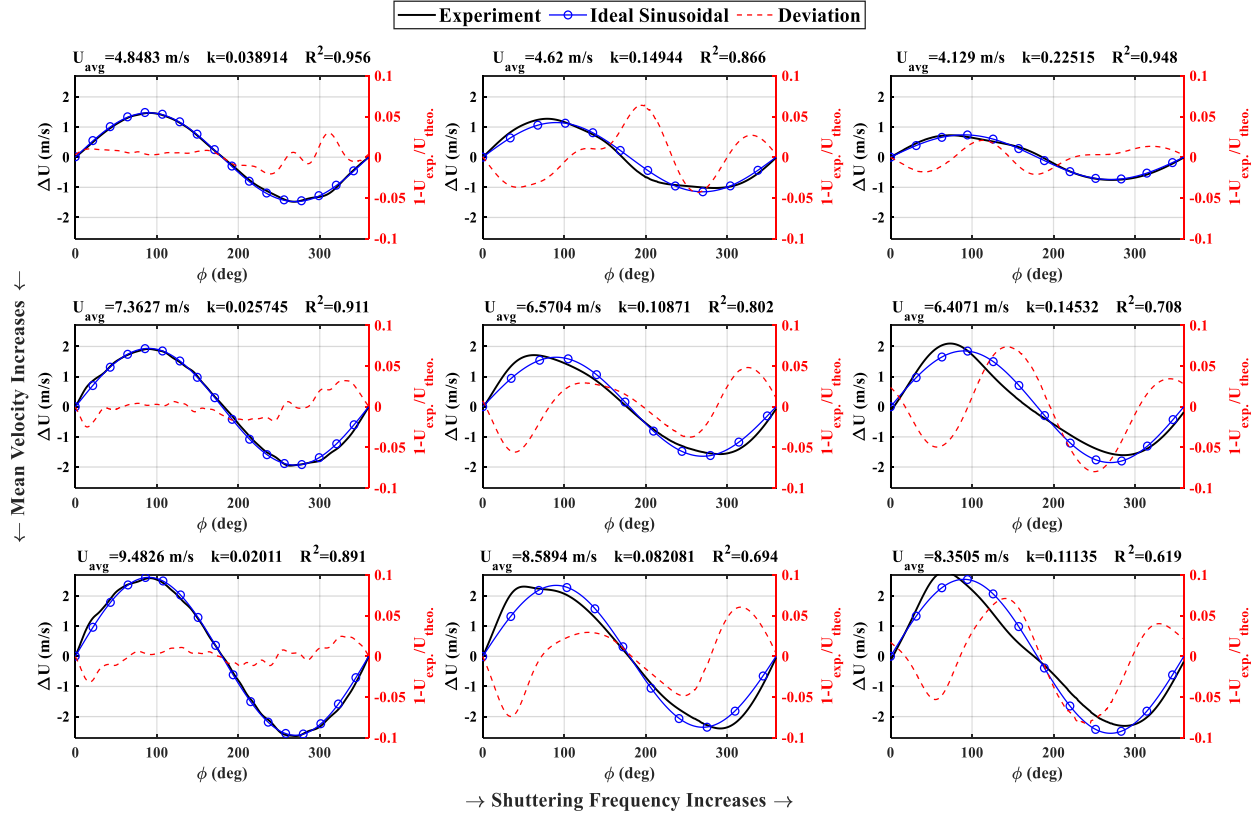


Figure 3. Experimental and theoretical velocity comparison and deviations. The rows represent cases with increase in mean velocity and the columns represent cases with increase in actuation frequency.

III. Results

A. Steady State Wing Response

The steady state coefficient of lift (C_L) and pitching moment (C_M) at different angles of attack will be discussed in this section. Figure 4a shows the variation of C_L with angle of attack for all of the configurations tested at 5 m/s ($Re = 50,000$). The configurations show the expected linear variation in C_L vs. α until each reaches its respective stall. The effective AR of the wing using the McCormick equation [20] shown in Equation 8 was found to be 2 as opposed to the geometric aspect ratio of 2.67.

$$C_{L,\alpha} = \frac{2\pi AR}{2 + \sqrt{4 + (AR)^2}} \quad (8)$$

It should be noted again that the propeller throttle setting was changed in each case to trim the wing, meaning to achieve a zero-drag condition. Since the wingtip and semi-midspan have two propellers, they operated at a lower RPM when compared to the midspan case to match the same thrust condition. The wingtip and midspan cases showed a small increment in $dC_L/d\alpha$ when compared to the baseline, but the semi-midspan case had the same slope as the baseline. The wingtip configuration has a higher $\alpha_{L=0}$ than the other two powered configurations. On the other hand, the unpowered model stalls at about 13° , with the powered configurations varying in stall angle and generally improving post-stall performance. The highest post-stall performer in terms of C_L is the midspan case, with the propeller positioned at the wing root, as no sharp stall behavior is present and a higher noticeable $dC_L/d\alpha$ is observed. Since the propeller was operating at a higher RPM compared to the semi-midspan and wingtip locations, the advance ratio of the propeller at midspan is lower than in the other cases. The propeller wake is dominant at lower advance ratios compared to the freestream and, as such, has a larger influence on the wing.

The difference between the powered and unpowered C_L performance is reduced at higher Reynolds numbers, as seen in Figure 4b and 4c, as the influence of propellers decrease at higher advance ratio. The unpowered wing

maintains the same performance, while the lift generated by the propeller has less of an effect at the higher U_∞ . The midspan configuration continues to extend and mitigate stall even while generating less lift at lower α . Wing performances in unsteady conditions at 5° and 15° angles of attack will be analyzed later in the paper.

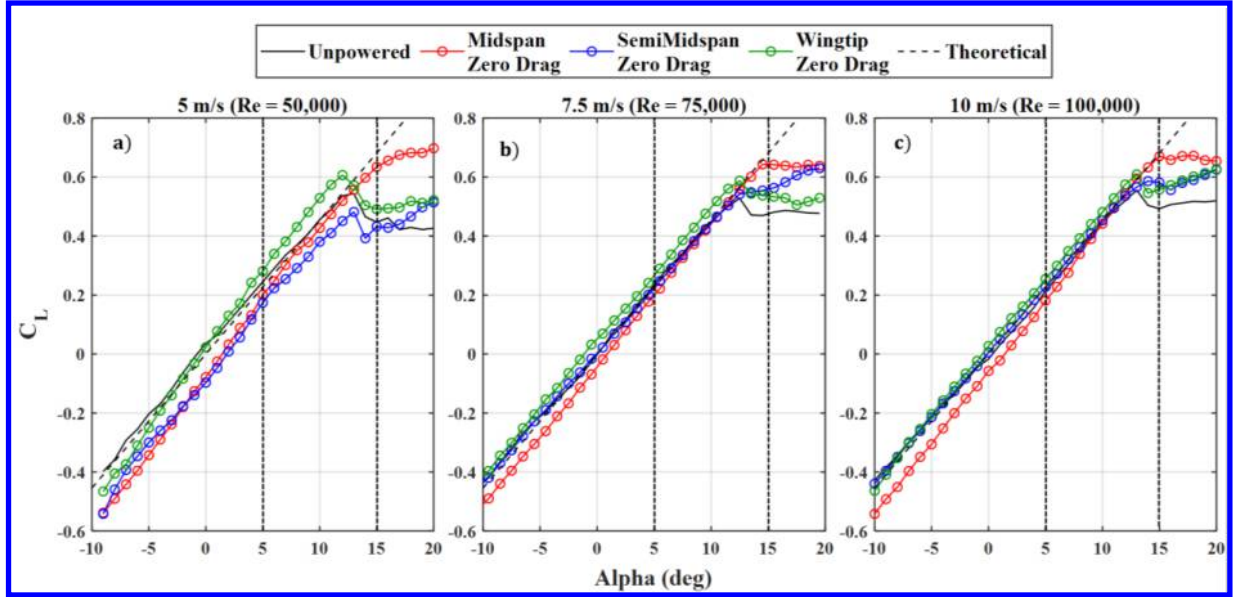


Figure 4. C_L vs. Alpha for Steady Flow Condition at different Reynolds numbers

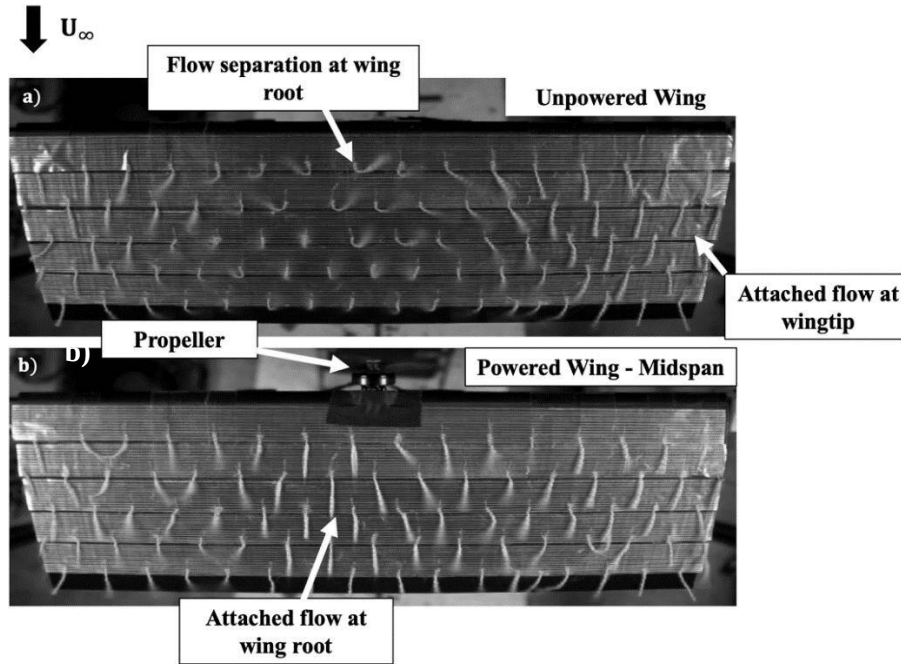


Figure 5. Tuft Flow Visualization at 15° and 10 m/s for the a) Unpowered and b) Midspan Configurations

Tuft flow visualization provides insight to the post-stall performance of the wing at 15° and 10 m/s. Figure 5a displays the Unpowered configuration, where blurred tufts are observed around the root area of the wing indicating flow separation. Meanwhile, the flow closer to the wingtip area remains attached due to the influence of the wingtip vortex. Figure 5b displays the midspan configuration with the propeller generating thrust. The propeller wake adds momentum to the flow over the wing and forces flow to be attached where it would normally be separated. This indicates the benefit of placing the propeller at the midspan of the wing closer to the stall condition, which agrees with the trend seen in Figure 4.

Figure 6 displays the coefficient of pitching moment between all configurations at different V_∞ . C_M is roughly constant and falls between ± 0.01 pre-stall while deviates highly post-stall, consistent with theoretical predictions for symmetrical wing. The differences between the unpowered and powered cases nearly overlap.

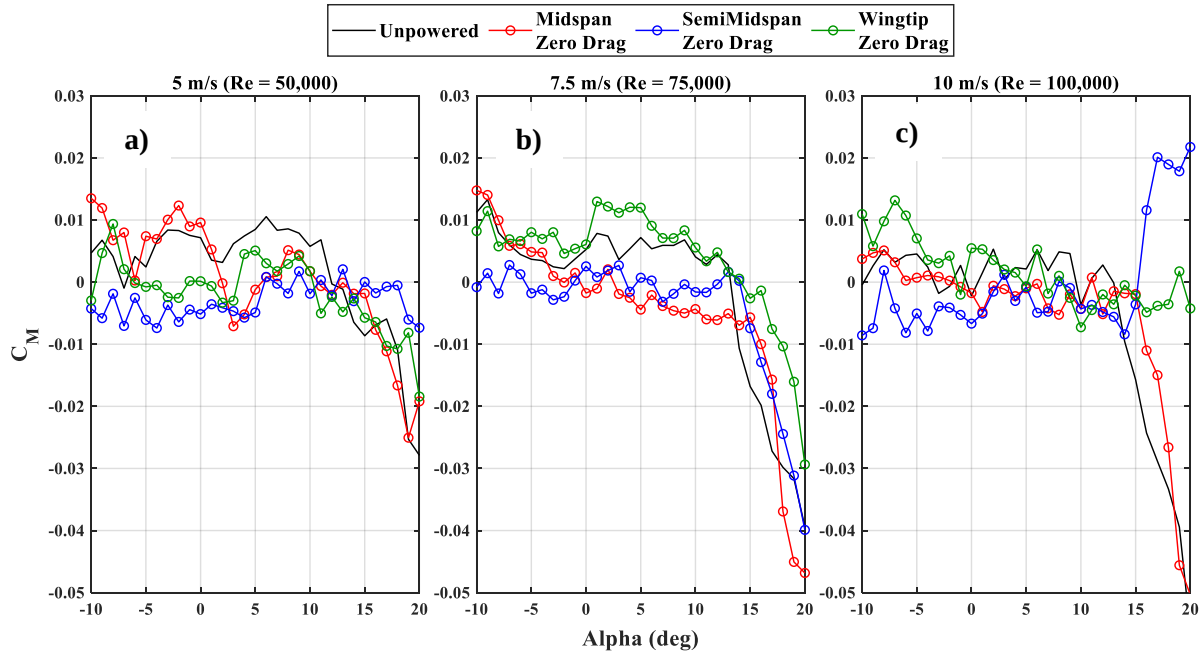


Figure 6. C_M vs. Alpha for Steady Flow Condition

B. Wing's Response to Periodic Streamwise Gusts

1. Quasi-steady response at 5° and 15° angles of attack

The quasi-steady regime of unsteady flow is considered any reduced frequency below 0.05, where the wing's response to small perturbations can be predicted by steady state models. Figure 7 contains the $C_L(\phi)$ of all configurations at the lowest reduced frequency tested, $k = 0.02$. Two methods are presented to visualize the response at 5° and 15° . The first method plots the normalized C_L as a function of ϕ through ~ 30 phase-averaged periods of the gust. At $\alpha = 5^\circ$ the unpowered C_L remains constant at the expected magnitude of ~ 0.2 from the steady state lift coefficient. At $\alpha = 15^\circ$, the unpowered configuration shows no changes with respect to phase, whereas the powered cases show an increase in lift at a phase angle of 270° where the velocity reaches its minimum. It is also worth noting that a higher C_L is reached at minimum U for the powered configuration, as the effect of a higher propeller thrust generation at lower advance ratio (or U_∞) as well as the less sensitivity to wind gust for the propeller at lower k [11]. The second method plots the C_L as a function of $\Delta U = U(\phi) - U_{avg}$. A full period of the gust completes a circle in this visualization. And any phase lag present in the response can be easily visualized through the hysteresis circle. This phase lag is inherent to unsteady flow and will increase as the reduced frequency increases. At 5° angle of attack, the unpowered and the powered wings don't show any hysteresis indicating that the response of the wings is truly quasi-steady. However, at a 15° angle of attack, there is a marginal phase lag between the velocity and C_L resulting in a slight hysteresis loop. This indicates that the powered wing is slightly susceptible to external disturbances when compared to the unpowered wing at higher angles of attack. And at these higher angles of attack, the unsteady threshold for powered configurations is lower than that for the unpowered configurations.

Figure 8 shows two similar visualization methods for $C_M(\phi)$ at both $\alpha = 5^\circ$ and 15° for the same configurations. At 5° angle of attack, the unpowered and powered configurations maintain a relatively constant C_M with ϕ and a minimal phase difference as evidenced in the hysteresis loop. At 15° angle of attack, the differences between the powered and unpowered wing in terms of magnitude and phase lag becomes significant, even at the very low reduced frequency considered. A significant hysteresis loop is evident at this high angle of attack indicating that the threshold for unsteady behavior for powered cases is lower than that of the unpowered cases.

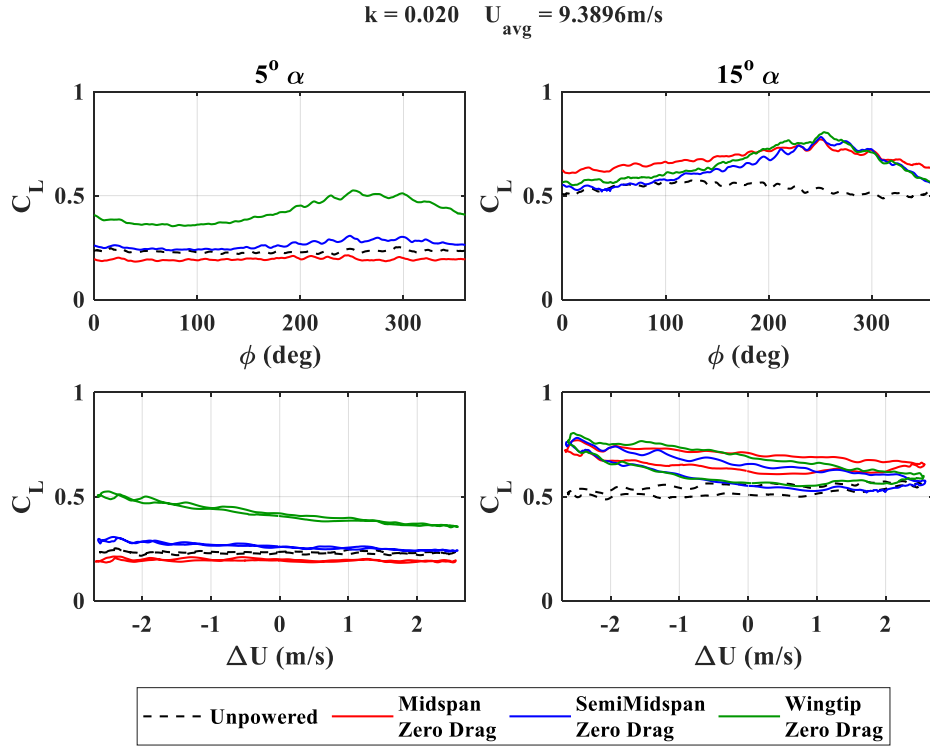


Figure 7. C_L vs. ϕ and C_L vs. ΔU for All Configurations

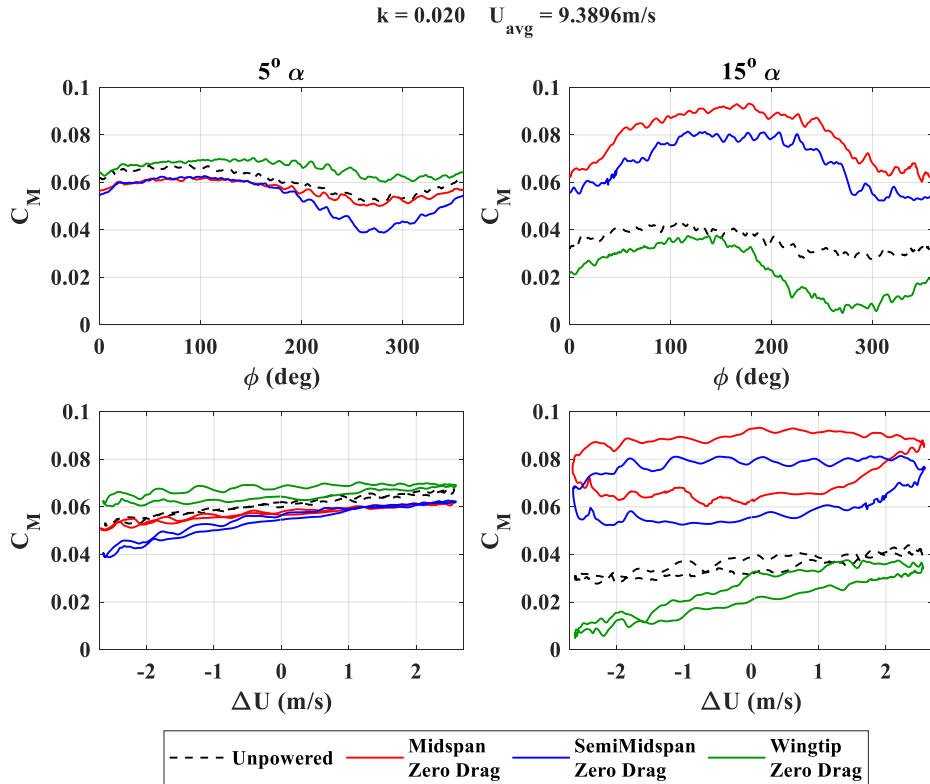


Figure 8. C_M vs. ϕ and C_M vs. ΔU for Unpowered and Midspan Configurations

2. Unsteady Response at 5° angle of attack

The wing's response to unsteady flow at a 5° angle of attack will be discussed in this section.. Figure 9 below displays the C_L as a function of phase angle ϕ at different reduced frequencies. The unsteady regime is traditionally considered within the reduced frequencies between 0.05 and 0.2 – the first row represents the cases with reduced frequency ranging from 0.08 to 0.11, and the second row represents the cases with reduced frequency ranging from 0.14 to 0.22. The response of the wings at these reduced frequencies is fundamentally different than the results seen in the quasi-steady case (Figure 7). At the lower reduced frequencies (0.082 - 0.11), all configurations follow the same general trend, where a small increment in C_L is observed when reducing U_∞ , then the change in C_L remains roughly constant as a function of phase angle until U_∞ reaches its minimum. A further increment in k results in a significant C_L peak at $\phi \sim 120^\circ$, as observed in Figure 9d. As the reduced frequency increases, the peak C_L formed shifts to higher ϕ as shown in Figure 9e. At the highest reduced frequency, the peak C_L occurs at $\phi \sim 230^\circ$. This trend in Figure 9f matches that of the results shown in Strangfeld [6]. The trends observed as a function of reduced frequency hold true for both powered and unpowered cases.

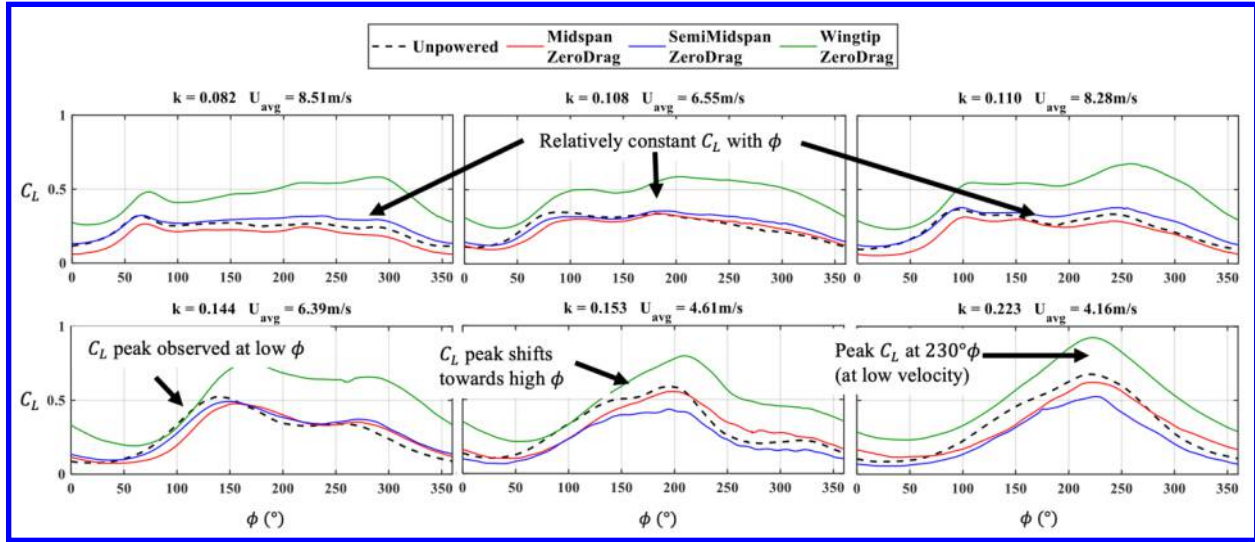


Figure 9. C_L vs. ϕ at 5° for cases at different reduced frequencies.

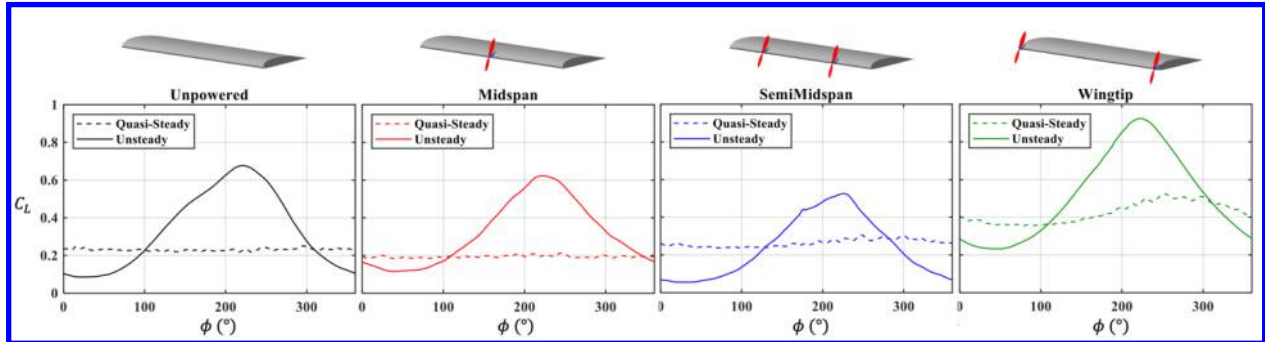


Figure 10. Quasi-Steady and Unsteady Response vs. ϕ at 5° for All Configurations

Recall from Figure 4, where the steady state C_L results at $\alpha = 5^\circ$ and 15° are different for each configuration and Reynolds numbers. These differences are replicated in the quasi-steady results (Figure 7) and the highest reduced frequency tested ($k = 0.22$) in Figure 10. To isolate the differences between each configuration due to the steady state performance and focus on the effect of unsteady aerodynamics, the quasi-steady aerodynamic performance (Figure 7) of each configuration is used as a reference and the C_L different between the steady and unsteady condition, ΔC_L , is calculated for all cases, which is defined as:

$$\Delta C_L(\phi) = C_{L,unsteady}(\phi) - C_{L,quasi-steady}(\phi) \quad (9)$$

Figure 11 recreates the results in Figure 9 in terms of ΔC_L . At low k (first row), ΔC_L for all configurations align quite closely with one another. As k increases (second row), the powered cases begin to deviate slightly from the unpowered case. While the midspan and wingtip configurations maintain their relationship with the baseline, a reduction of C_L from the semi-midspan configuration is observed. This configuration has the highest propeller coverage along the leading edge of the wing when compared to other cases, which results in a smaller magnitude of ΔC_L under gust encounter, as the propeller wake is less sensitive to the changes in U_∞ under the advance ratio tested when holding a constant rotational speed [11]. Therefore, a lower change in velocity across the span of the wing.

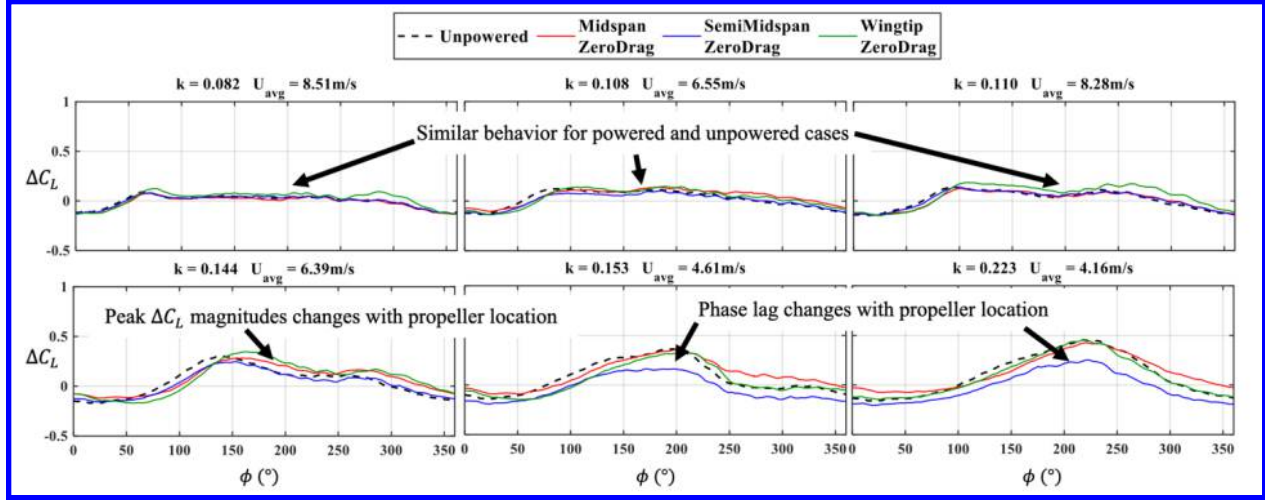


Figure 11. ΔC_L vs. ϕ at 5° for all configurations

The C_L trends with respect to the reduced frequency annotated in Figure 9 are clearly seen in Figure 11. Again, the phase angle ϕ , at which the ΔC_L peak occurs, differs marginally for different powered cases. The changes in phase lag for the various powered cases are distinctly visible in Figure 12. At low reduced frequencies (first row), the responses of all powered wings coincide with the unpowered wing's response, indicating no differences in phase lag. However, at higher reduced frequencies, slight variations in the hysteresis loop are observed due to the changes in the ϕ location at which peak ΔC_L occurs. In all the cases, the peak C_L leads the velocity in phase.

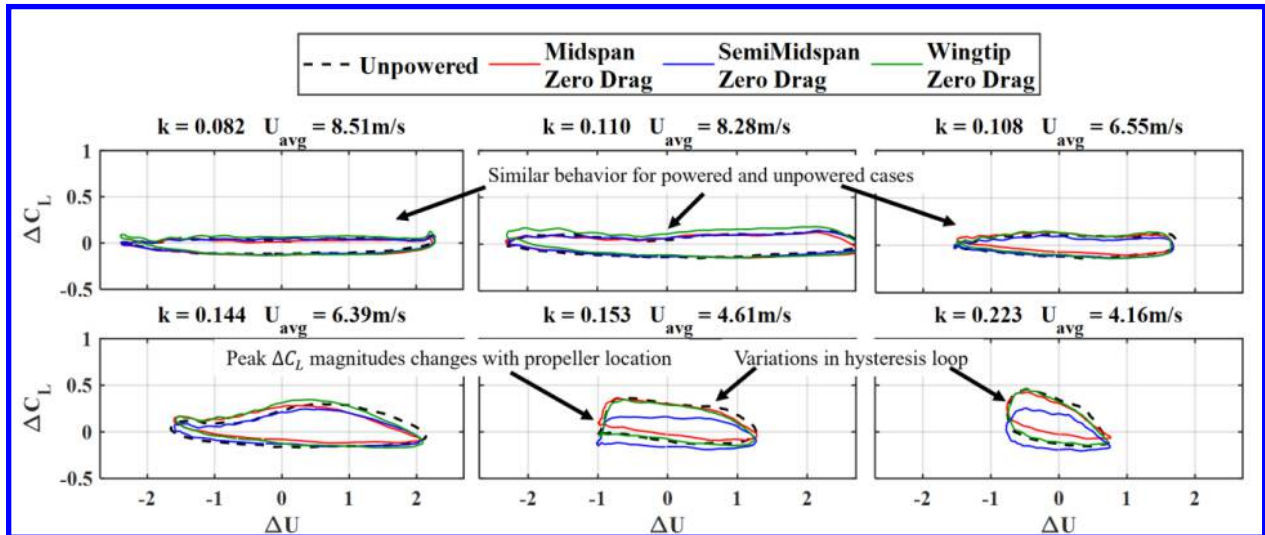


Figure 12. ΔC_L vs. ΔU at 5° for All Configurations

Figure 13 displays the ΔC_M as a function of ϕ to characterize the pitch stability of a powered wing in unsteady flow. When comparing Figure 13 with Figure 11, ΔC_M almost completely matches the general trends seen in the corresponding ΔC_L , where the changes in ΔC_M is almost identical at smaller k and deviates between different

configurations at higher k . It is also worth noting that the semi-midspan and wingtip configurations create a larger ΔC_M than the midspan configuration, indicating a reduction in pitching moment stability for these configurations in unsteady flow. The ΔC_M phase lag between the powered cases also follows the same trend as ΔC_L , except at the highest reduced frequency. At $k = 0.22$, the midspan case achieves the peak ΔC_M at earlier ϕ when compared to the other cases. This change in phase lag in the midspan case is evident in the ΔC_M vs ΔU plot shown in Figure 14. The different responses in ΔC_L and ΔC_M phase lag threatens the flight stability at the highest reduced frequency, especially for aircraft undergoing transition from vertical to horizontal flight. However, at lower reduced frequencies, this is not a concern.

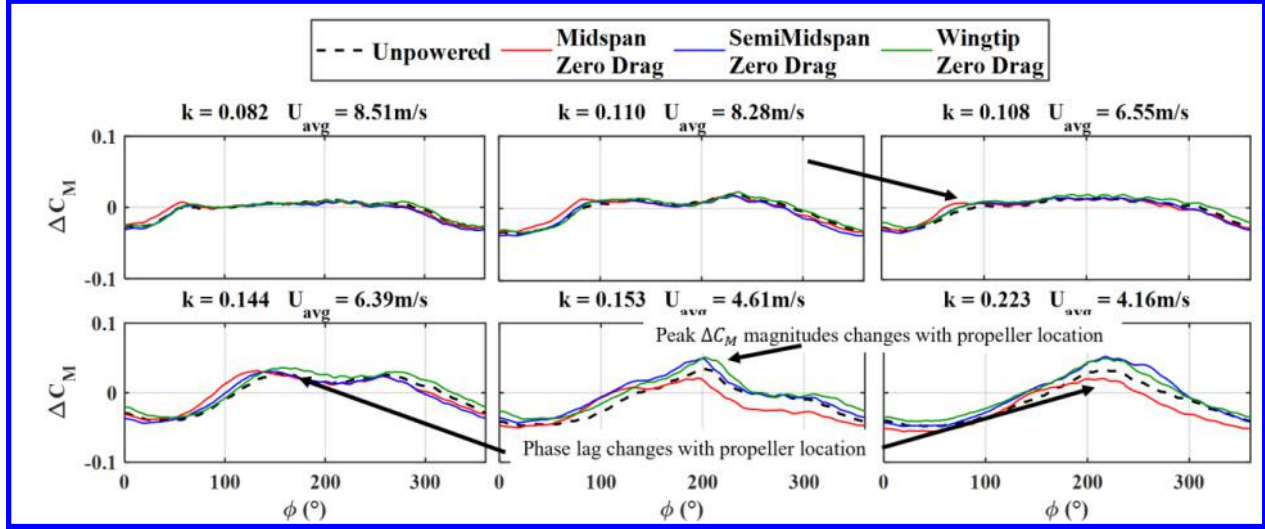


Figure 13. ΔC_M vs. ϕ at 5° for All Configurations

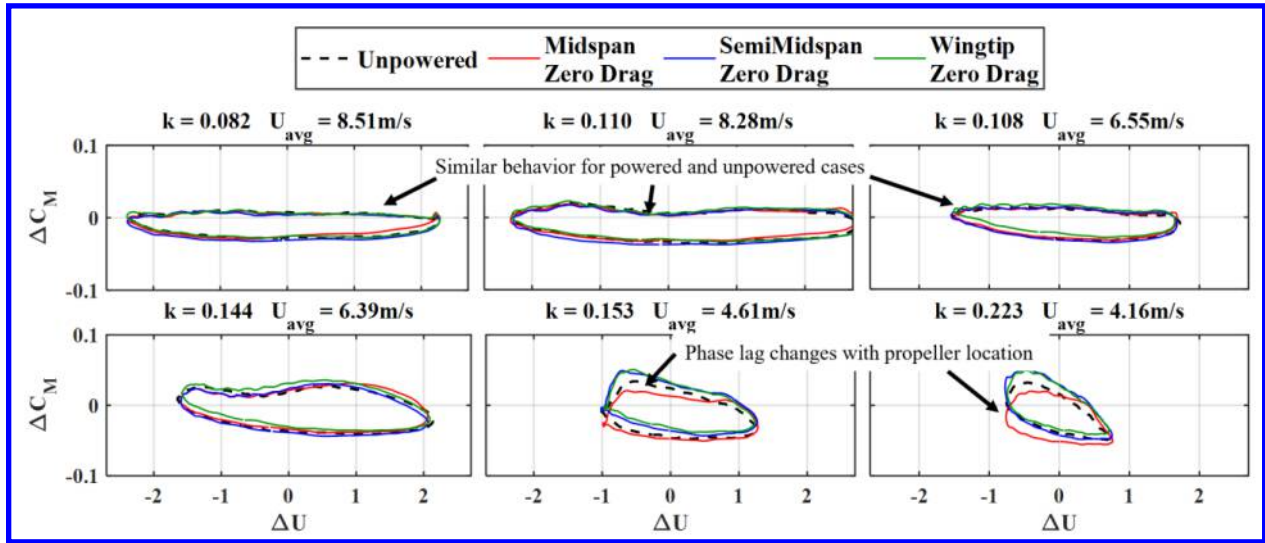


Figure 14. ΔC_M vs. ΔU at 5° for All Configurations

3. Unsteady Response at 15° angle of attack

The unsteady response of all configurations at $\alpha = 15^\circ$ is shown in Figure 15. The ΔC_L post-stall, where flow separation occurs, displays much higher variability between configurations than pre-stall (Figure 11), particularly in the midspan cases. In general, the unpowered, semi-midspan, and wingtip configurations perform similarly in terms of trend and magnitude. The phase ϕ at which the ΔC_L peak locations occur is very similar to the trends seen at 5° angle of attack. Again, the midspan configuration achieves its peak ΔC_L at a lower ϕ when compared to the other cases

at higher k as annotated. At these reduced frequencies, the midspan case also has a higher ΔC_L than any other configurations, as a result of a higher propeller-wing interaction for this configuration at post stall.

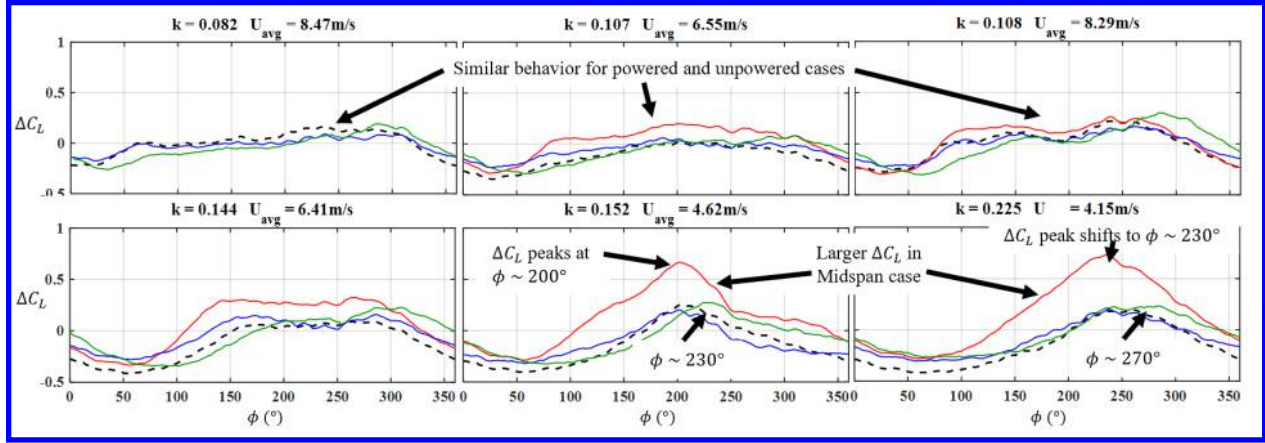


Figure 15. ΔC_L vs. ϕ at 15° for All Configurations

Figure 16 replots the results in Figure 15 against ΔU . The size of the hysteresis loop increases with the increase in k . The larger magnitude of ΔC_L and the lower ϕ at which the peak ΔC_L occurs makes the midspan hysteresis loop much larger when compared to other powered cases at higher reduced frequencies.

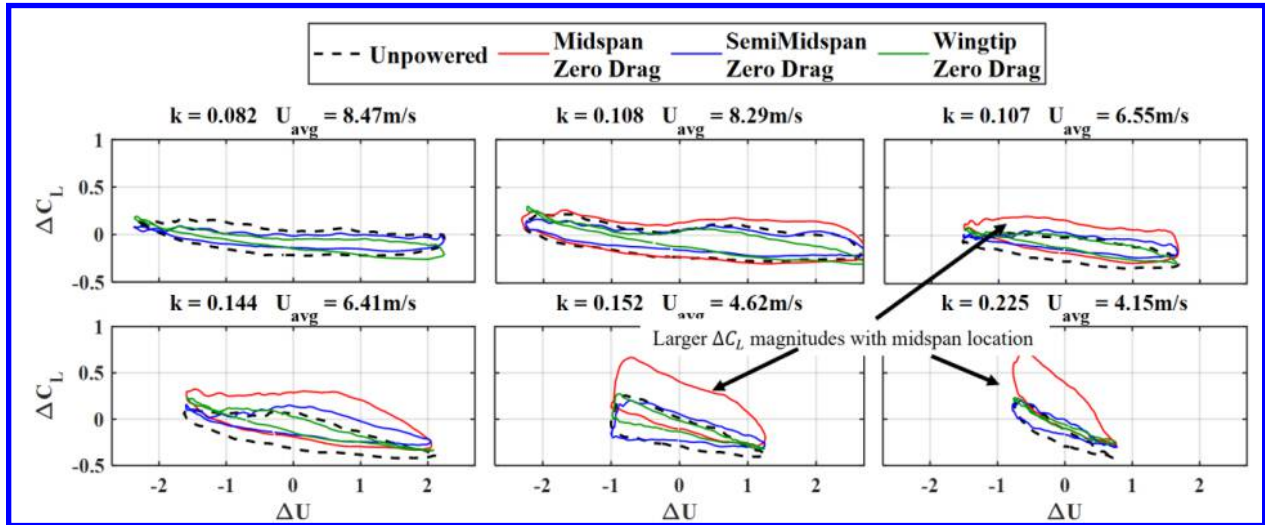


Figure 16. ΔC_L vs. ΔU at 15° for All Configurations

According to the tuft flow visualization for the unpowered and midspan configurations in unsteady flow shown in Figure 16, the flow separation occurs at the lowest U_∞ at ($\phi = 270^\circ$) for the unpowered wing. The flow reattaches when increasing U_∞ and stays attached until the next period. When placing the propeller at midspan, the additional momentum from the propeller forces flow attachment through the normally stalled region of the wing at $\phi = 270^\circ$. The addition of the propeller momentum has a much smaller effect at the semi-midspan and the wingtip case than the midspan case since the flow remains partially attached at these locations even in the unpowered cases.

The post-stall ΔC_M vs. ϕ is shown in Figure 18. The variability between configurations increases greatly when compared to the $\alpha = 5^\circ$ condition (Figure 14), which is to be expected with flow separation occurs at stall. The added benefits from flow attachment in the midspan configuration led to a lower variation in ΔC_M through ϕ when compared to the other powered configurations, indicating a positive effect of placing the propeller at mid-span for stabilizing the aircraft at post stall conditions.

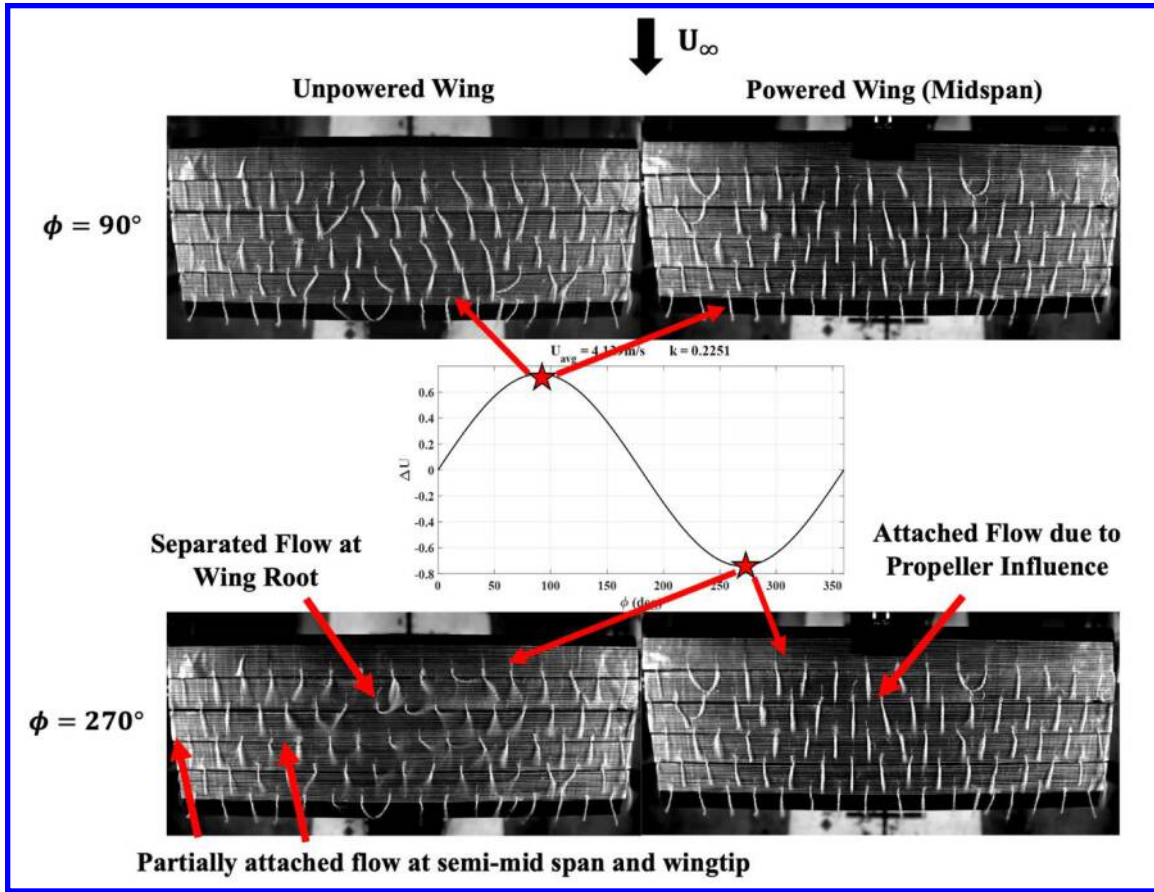


Figure 17. Tuft Flow Visualization at 15°, $k=0.23$, Unpowered (left) and Midspan (right)

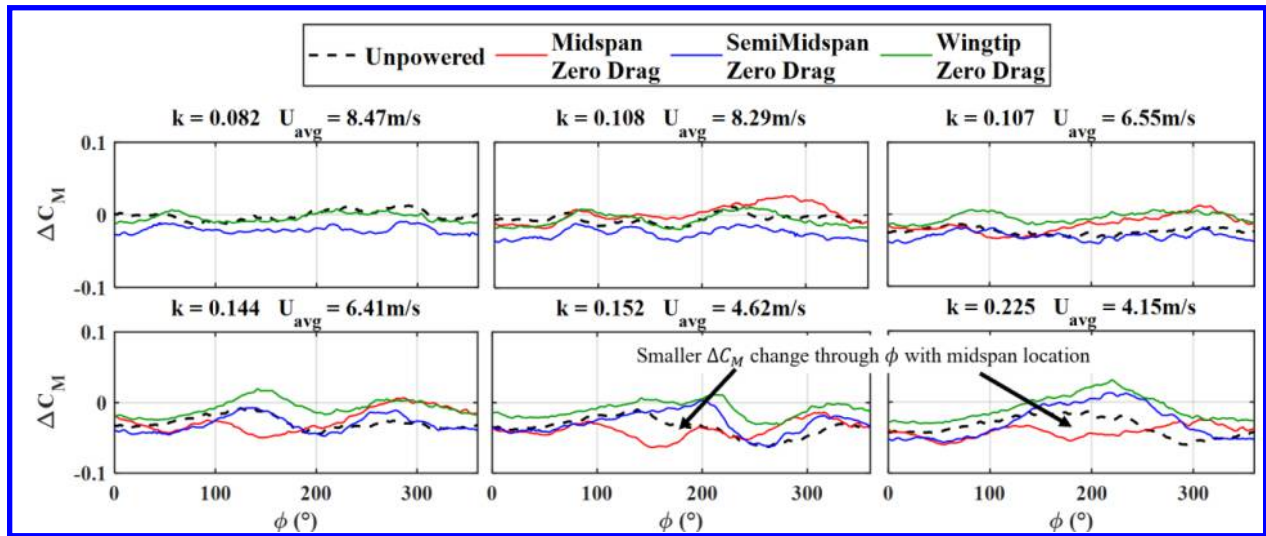


Figure 18. ΔC_M vs. ϕ at 15° for All Configurations

Figure 19 displays the ΔC_M as a function of ΔU at $\alpha = 15^\circ$. From pre-stall to post-stall, the hysteresis loop collapses for all reduced frequencies, except for the highest k tested, where the mid-span configuration shows a smaller hysteresis loop than other configurations. Moreover, at the highest k , the phase lag is diminished when compared to the 5° condition for all cases. The ΔC_M can have the same magnitude at multiple points along the period.

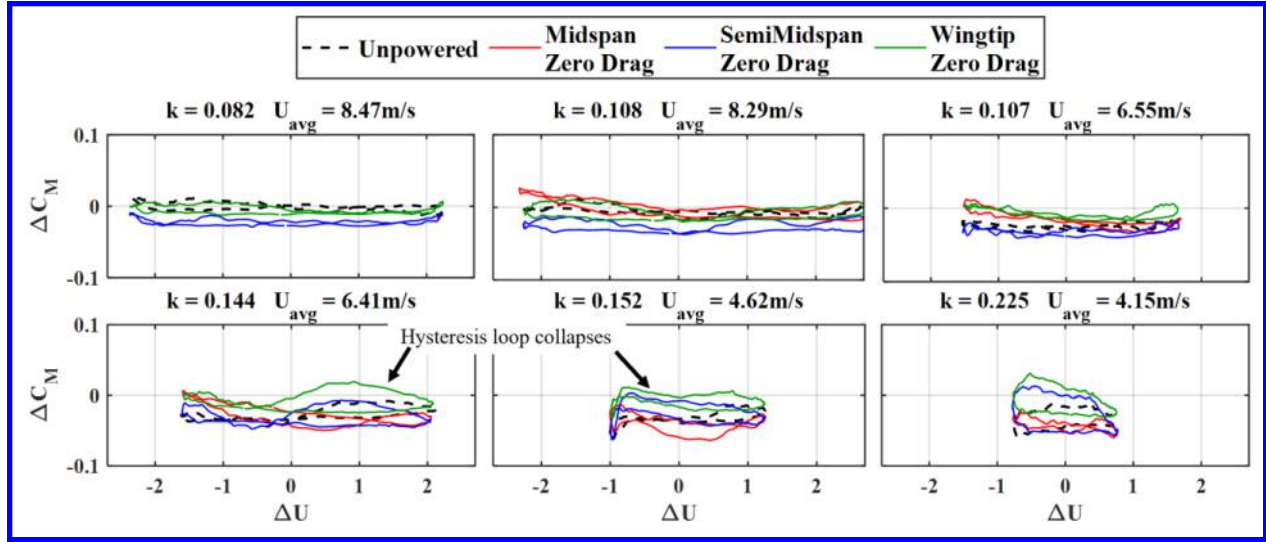


Figure 19. ΔC_M vs. ΔU at 15° for All Configurations

IV. Conclusion

The current study investigated unsteady effects on a propeller-wing system at different reduced frequencies and amplitudes. Three configurations featuring different propeller locations (Midspan, Semi-Midspan, and Wingtip) were tested, along with an unpowered wing at two different angles of attack. The propeller in the midspan configuration operated at a lower advance ratio (higher RPM), while the propellers in the semi-midspan and wingtip configurations operated at a relatively higher advance ratio (lower RPM) to ensure a trimmed condition (zero drag). The behavior of the powered and the unpowered wings across different reduced frequency range is summarized below:

- In quasi-steady state, at a 5° angle of attack, the differences between powered and unpowered lift and moment coefficients were negligible. However, at a 15° angle of attack, a phase lag between velocity and ΔC_L was observed, indicating that the threshold for unsteady behavior for powered cases is lower than that in unpowered cases, especially at higher angles of attack.
- At moderate reduced frequencies ($k = 0.08 - 0.11$), ΔC_L remains relatively constant as a function of phase angle ϕ . An increase in k resulted in a significant C_L peak to appear at low ϕ of 150° . As the reduced frequency increases, the peak ΔC_L formed shifts to a higher ϕ of 230° . This trend holds for all powered and unpowered cases at both 5° and 15° angles of attack. However, for the midspan cases at a 5° angle of attack, the ΔC_L and ΔC_M phase lag do not follow the same trend.
- A similar trend is observed at higher reduced frequencies ($k = 0.15 - 0.22$), but the differences among powered wing cases become more prominent. At 15° angle of attack, the midspan case exhibits a higher ΔC_L than any other configurations, as a result of increased propeller-wing interaction for this configuration at post stall. ΔC_L and ΔC_M leads the velocity at different ϕ resulting in increased sensitivity to propeller location.

Overall, it can be concluded that the powered wing's response to unsteadiness is more pronounced at high angles of attack and higher reduced frequencies. The location of the propeller and the advance ratio at which it operates can alter the wing's response to unsteadiness.

Acknowledgments

The author would like to thank the University of Dayton Graduate Student Summer Fellowship and the Department of Mechanical and Aerospace Engineering for their support throughout this process. The author would also like to thank Michael Mongin for his insight and advice on this study.

References

- [1] Palaia G, Abu Salem K, Cipolla V, Binante V, Zanetti D. “A Conceptual Design Methodology for e-VTOL Aircraft for Urban Air Mobility.” *Applied Sciences*. 2021; 11(22):10815. <https://doi.org/10.3390/app112210815>
- [2] Hostenbach, M., Nottenoom, C., Theys, B., De Schutter, J., “Design and Control of an Unmanned Aerial Vehicle for Autonomous Parcel Delivery with Transition from Vertical Take-off to Forward Flight – VertiKUL, a Quadcopter Tailsitter”, *International Journal of Micro Air Vehicles*, 7 (4), pp. 395–405, 2014. <https://doi.org/10.1260/1756829374395>
- [3] Barth, J., Condomines, J.P., Bronz, M., Moschetta, J.M., Join, C., Fliess, M., “Model-Free Control Algorithms for Micro Air Vehicles with Transitioning Capabilities”, *International Journal of Micro Air Vehicles*, 12 (1), pp. 1-12 2020. <https://doi.org/10.1177/1756829320914264>
- [4] Stone, R.H., Anderson, P., Hutchinson, C., Tsai, A., Gibbens, P., Wong, K.C., “Flight Testing of the T-Wing Tail-Sitter Unmanned Air Vehicle”, *Journal of Aircraft*, 45 (2), pp. 673-685, 2008. <https://doi.org/10.2514/1.32750>
- [5] Jones, A., Cetiner, O., Smith, M.J., “Physics and Modeling of Large Flow Disturbances: Discrete Gust Encounters for Modern Air Vehicles”, *Annual Review Fluid Mechanics*, 54 (1), pp. 469-493, 2022. <https://doi.org/10.1146/annurev-fluid-031621-085520>
- [6] Isaacs, R., "Airfoil Theory for Flows of Variable Velocity," *Journal of the Aeronautical Sciences*, 12 (1), pp. 113-117, 1945.
- [7] Strangfeld, C., Müller-Vahl, H., Nayeri, C.N., Paschereit, C.O., Greenblatt, D., “Airfoil in a High Amplitude Oscillating Stream”, *Journal of Fluid Mechanics*, 793 (1), pp. 79-108, 2016
- [8] Zhu, W., McCrink, M.H., Bons, J.P., Gregory, J.W., “The Unsteady Kutta Condition on an Airfoil in a Surging Flow”, *Journal of Fluid Mechanics*, 893 (2), pp. 1-12, 2020.
- [9] Leishman, J. G., “Principles of Helicopter Aerodynamics”, Cambridge University Press, 2006.
- [10] Lowey, R. G., “A Two-Dimensional Approximation to the Unsteady Aerodynamics of Rotary Wings”, *Journal of the American Helicopter Society*, 24 (2), pp. 81-92, 1957.
- [11] Cai, J., Gunasekaran, S., “Frequency Response of RC Propellers to Streamwise Gusts in Forward Flight”, *MDPI-Wind*, Vol.3(2), pp. 253-272, 2023, doi: 10.3390/wind3020015
- [12] Wang, B., Ali, Z. A., and Wang, D., “Controller for UAV to Oppose Different Kinds of Wind in the Environment” *Journal of Control Science and Engineering*, Vol. 2020., ID. 5708970, 2020
- [13] Whidborne, J. F., and Cooke, A. K., “Gust Rejection Properties of VTOL Multirotor Aircraft”, 2017 IFAC, 50-2 (2017), pp. 175-180
- [14] “What are the Advantages and Disadvantages of Turboprop Engines?”, National Aviation Academy, Retrieved from: <https://www.naa.edu/>
- [15] Aref, P., Ghoreyshi, M., Jirasek, A., Satchell, M.J., Bergeron, K., “Computational Study of Propeller-Wing Aerodynamic Interaction”, *Aerospace*, 5 (3), pp. 79-99, 2018.
- [16] Srivathsan, S., Rauleder, J., “Experimental Wind Tunnel Investigation on Propeller-Wing Interactional Aerodynamics”, *AIAA SCITECH Forum*, 2023.

- [17] Cook, Thomas, Sidaard Gunasekaran, Michael V. Ol, and Michael P. Mongin. "Frequency Response of a Shuttered Open Jet Wind Tunnel." In AIAA Scitech 2020 Forum, p. 1760. 2020
- [18] DANTEC Dynamics, Hot wire and Mini-CTA 54T30, Retrieved from: <https://www.dantecdynamics.com>
- [19] ATI, F/T Mini-40 Sensor, Retrieved from: <http://www.ati-ia.com/>
- [20] McCormick, B. W., "Aerodynamics, Aeronautics, and Flight Mechanics (Second Edition)," John Wiley & Sons, New York, 1995, p. 116.